

31338 Network Servers

Lab 5b: Superuser powers

Aims

1. To learn how to safely use root/admin access
2. To notify users about system issues or company computer usage policies

Task 1: Executing root commands safely via sudo in Linux

By now on Linux you would have discovered the **su** command which creates a shell owned by **root** or simply logged on as root to execute privileged commands

This is extremely dangerous since any command can be executed, and there is no real audit of who used the root account.

The best practice is to make use of the **sudo** command to execute specific commands.

For this exercise, we will allow users to execute the `ifconfig` command with administration privileges

From root, execute the command

```
visudo
```

 (all one word, no space)

This edits the `/etc/sudoers` file, which is normally only `rw` to root.

Note: if you really cannot use `vi` you can make `visudo` use a different editor. First run:
`export EDITOR=gedit`
and then run `visudo` afterwards (or any other plain text editor works too, e.g. `nano`).

The format of the file is not very obvious (hint: `man 5 sudoers`) but generally it defines users and the set of commands that the user may run, optionally limiting to particular machines.

```
USER    [MACHINE=]COMMAND
```

Look through the file that `visudo` displays. What commented-out user permissions are there? These are standard permissions a system administrator might wish to grant.

Read the following section:

```
## Allows members of the users group to mount and unmount the
## cdrom as root
# %users    ALL=/sbin/mount /mnt/cdrom, /sbin/umount /mnt/cdrom
```

Why might this command be necessary? Why does it only match `/mnt/cdrom`, and not any device?

If an administrator wanted to grant all users the ability to disconnect or reconnect the `ens33` network port, (`ifconfig ens33 up`, `ifconfig ens33 down`), how should this be written to `visudo`? Do this.

Save and close out of `visudo`, and make sure that your normal user (peter, lois, etc) is able to run `'sudo ifconfig ens33 down'` and `'sudo ifconfig ens33 up'` but **not** `'sudo tail /etc/shadow'`.

Execute `visudo` again as root, remove that rule, and grant the same `sudo` permissions to your user as the root user. (Copy root's line, using your own username) Save and quit, and test that you can run any command as root using `'sudo <cmd>'`.

Finally, what happens if you run (while logged in as a normal user): `sudo su -`

What does it do? Is this a good thing? Why/why not? Make notes in your journal about all of the above steps.

Task 2: Executing admin commands in Windows Server

Windows UAC (User Account Control) will control whether any command you execute will need admin authority and will prompt you whether to do so. You can usually execute a command with administrator rights by using Right Mouse Button → Run As Administrator in the GUI.

However, sometimes you need to run as another user. You can use Right Mouse Button → Run As ... then pick the user (and you will need the password), but you can also do this from the command line:

```
runas /user:userid program
```

Note that userid can also be in a domain (domain\userid) format

While logged in as a normal user (peter, lois, etc), try executing the `cmd.exe` shell as the Administrator:

```
runas /user:Administrator cmd.exe
```

Notice that the label of the window is now ‘Administrator’ and you can view any directory on the system.

Now log out of your normal user account and log back in with the Administrator account. Try running `cmd.exe` as one of your normal users, e.g. peter:

```
runas /user:peter cmd.exe
```

What do you notice? How is this different to Linux?

(e.g. in Linux if you are logged in as root and type ‘`su peter`’, what happens?)

Task 3: Notifying users on system issues or computer usage policies (Linux – the old way)

Edit the file `/etc/motd` (message-of-the-day) and put in a message. Log out and log back in. When is this message displayed? Now use `ssh` to connect to your local machine as your normal user. You can use either of the following style of commands:

```
ssh your_username@localhost
ssh -l your_username localhost
```

(Note that the option in the second form is “dash lower-case L”).

Is the `/etc/motd` file displayed for `ssh` sessions? Remember to exit from your `ssh` session.

Now, press Ctrl-Alt-F3 (hold down the Ctrl + Alt keys and press F3). Or F4. Or F5. Each of these is a different console login (text-based) that is available. Try logging in now as one of your users using the text login approach. Is `/etc/motd` displayed? When?

To go back to the graphical screen, press Ctrl-Alt-F1.

Now put some kind of welcome message inside the file `/etc/issue`. Put something different than in `/etc/motd`.

Let’s see when (and if) these are used. Try:

- Graphical login (Log out and log in again. Is `/etc/issue` displayed? Is `/etc/motd` displayed? When?)
- Text-based login by pressing Ctrl-Alt-F3 (Is `/etc/issue` displayed? Is `/etc/motd` displayed? When?)
- Also try connecting with `ssh` (Is `/etc/issue` displayed? Is `/etc/motd` displayed? When?)

There is also `/etc/issue.net`! Try and work out when this one is shown (hint: you probably need to install a telnet server and telnet client). You can also use the “banner” option in `/etc/ssh/sshd_config` to show `/etc/issue.net`

Task 4: Notifying users on system issues or computer usage policies (Linux – the graphical way)

/etc/motd and /etc/issue are part of a very old system and it was only ever designed for text logins. Modern workstations have graphical logins. If we want a message on the graphical login screen (gdm), there's a few steps:

1. Create a new file **/etc/dconf/profile/gdm** – or edit if it already exists – with the contents:

```
user-db:user
system-db:gdm
file-db:/usr/share/gdm/greeter-dconf-defaults
```

2. Create a second new file **/etc/dconf/db/gdm.d/01-banner-message** with the contents:

```
[org/gnome/login-screen]
banner-message-enable=true
banner-message-text='Greetings wonderful Linux user!'
```

3. Update the dconf database by running the command:
`dconf update`
4. Restart the gdm service (Gnome Display Manager) by running the command:
`systemctl restart gdm`

Note: you can also put an image/logo on the login screen. If you like, you can research how to do this (optional). The process is quite similar to what we did above.

Task 5: Notifying users on system issues or computer usage policies (Windows Server)

On Windows Server, you can similarly display a message-of-the-day or welcome message or company security policy message on the login screen.

1. Go to Server Manager → Tools → Local Security Policy
2. From the list of security settings on the left, navigate to Local Policies → Security Options. Scroll down and look for the entries that start with “Interactive logon”.
3. We need to edit two of them. To edit, double click on the name and a window should open for you to enter text. The two we need to edit are:
 - Interactive logon: Message **text** for users attempting to log on
 - Interactive logon: Message **title** for users attempting to log on

The message title appears at the top and is just a single line (a few words). The message text can be longer. Save your changes.

4. Open a Command Prompt or PowerShell window and run:

```
gpupdate /force
```

This updates the system policies.

5. To test, log out and log back in again. Your message should be shown.